

AIGR Gene Hypothesis Deep Research — Horse SHLD2 (A0A9L0RGD6) and NHEJ Promotion

Focus: function_assignment · **Hypothesis slug:** horse40-nhej-promotion **Target GO term:** GO:2001034 — *positive regulation of double-strand break repair via nonhomologous end joining*
Target protein: Equus caballus SHLD2, UniProt A0A9L0RGD6 (883 aa) **Human comparison lead:** SHLD2, UniProt Q86V20 (835 aa)

Summary

Verdict: Supported by justified mammalian (orthology) transfer — not by direct horse evidence. The horse protein A0A9L0RGD6 is an unambiguous ortholog of human SHLD2 (Shieldin complex subunit 2), the single-stranded-DNA-binding subunit of the Shieldin complex that acts downstream of 53BP1–RIF1 to restrict DNA end resection and thereby promote nonhomologous end joining (NHEJ). A banded global alignment of the frozen 883-residue horse sequence against human Q86V20 gives 72.6% identity over 793 aligned columns, rising to **86.3% across the OB1/OB2 ssDNA-binding core** and holding at ~70% across the N-terminal SHLD3/REV7 recruitment segment. Because the exact target GO term (GO:2001034) is a curated **IDA** (Inferred from Direct Assay) annotation of human SHLD2, and the molecular apparatus underlying that annotation — the OB-fold ssDNA-binding module plus the N-terminal recruitment motif — is conserved and structurally intact in the horse protein, the hypothesis is **well justified as a homology-based transfer**. The appropriate evidence code is **ISS or IBA (orthology/phylogenetic inference), not experimental (IDA/EXP)**, because no horse-native functional assay exists.

Two caveats are important for a curator. First, **all support is by mammalian transfer**; there is no directly observed Equus caballus experimental evidence for SHLD2 in DSB repair. A PubMed search returned no horse-native SHLD2 functional study. Second, the **frozen UniProt sequence carries a C-terminal frameshift** that corrupts the third OB fold (OB3 / SHLD2_C). Perfect colinear identity with human SHLD2 extends to horse residue ~803 (shared motif ...FTMKKIYY), after which the horse sequence diverges into a spurious, arginine/glycine/proline-

rich tail (45% R+G+P versus 11.7% genome-wide) — the canonical compositional signature of an out-of-frame translation. This is almost certainly a gene-model / prediction artifact rather than real horse biology. Critically, the frameshift leaves OB1 and OB2 — the domains that execute ssDNA binding and resection blockade — intact, and AlphaFold confirms both fold with high confidence.

For the curator, the recommended lead is: **assign GO:2001034 to horse SHLD2 with an ISS or IBA evidence code** (reference ortholog human Q86V20), **not IDA/EXP**, and flag the OB3 frameshift as a sequence-model issue to verify against a corrected gene model. The seed hypothesis is neither refuted nor experimentally confirmed for horse; it is a defensible orthology-based function assignment.

Key Findings

F001 — Horse A0A9L0RGD6 is a bona fide SHLD2 ortholog with conserved recruitment and core ssDNA-binding apparatus

A banded global alignment (identity scoring, band 180) of the frozen 883-aa horse sequence (length verified = 883; SHA-256 as supplied) against human SHLD2 (Q86V20, 835 aa) yielded **72.6% identity over 793 aligned columns**. Resolving identity by functional region against human coordinates gives a clear picture: the N-terminal disordered region (human 1–420) is 67.1% identical (282/420); the **OB1/OB2 core (human 421–720) is 86.3% identical (259/300)**; and the C-terminal OB3 region (human 721–835) drops to 30.4% (35/115) — the signal of the frameshift described in F002. The functionally decisive **SHLD3/REV7-binding N-terminal segment (human 1–60) is 70% identical (42/60) and colinear**, and the horse protein conserves the exact functional motif `SLPNIVYTGCAKCGLELETDENRIY...RQCFSCLPFTMKKIYY`.

This architecture matches the defining SHLD2 domain layout — an N-terminal intrinsically disordered region followed by three C-terminal OB folds homologous to RPA1 — reported by [PMID: 32306447](#): *"FAM35A has unique domains: an N-terminal disordered domain and three C-terminal OB-fold domains. These C-terminal domains have homology with the OB-fold domains of the single-stranded DNA binding protein, RPA1. With other 53BP1-pathway factors, FAM35A inhibits DNA end resection."* The near-perfect conservation of both the recruitment segment and the ssDNA-binding OB1/OB2 core establishes that the horse protein possesses the molecular apparatus required for the Shieldin-mediated NHEJ-promoting role — the crux of

the decisive question. Identity, reciprocal domain architecture, and the PANTHER subfamily assignment all point specifically to SHLD2, ruling out paralog confusion with RPA1 or other OB-fold proteins.

F002 — The frozen horse SHLD2 UniProt record contains a C-terminal frameshift that corrupts OB3

Alignment shows **perfect colinear identity up to horse residue 803 / human residue 730** (shared motif ...FTMKKIYY), after which the horse sequence diverges completely. Human SHLD2 continues `RPALMTAIDGRHDVCIRVESKLIK...` (canonical OB3), whereas the horse record continues `SAPRRGHLWHGRRRLAPRLAGGRRGPLCGE...`. The 80-residue horse C-terminal tail is **45% Arg+Gly+Pro versus 11.7% genome-wide** in human SHLD2 — the compositional signature of an out-of-frame reading frame. The horse protein is also 48 residues longer than human (883 vs 835), consistent with a frameshift extending translation past the true stop codon into a spurious frame.

The interpretation is that this is a **gene-model / automated-prediction artifact**, not a biologically meaningful divergence in the horse lineage. A corrected RefSeq/Ensembl model would likely restore an intact OB3. OB3 contributes to SHLD1 binding and complex stability but is not the primary ssDNA-binding execution module; its corruption therefore does not, on its own, negate the resection-restriction function that OB1/OB2 supply. A curator should treat the frozen sequence as provisional and recommend verification against a corrected transcript or genome assembly.

F003 — SHLD2's NHEJ-promoting role is experimentally established in human/mouse (the basis for horse transfer)

Primary literature in human and mouse systems establishes SHLD2 as the **ssDNA-binding subunit of the Shieldin complex that acts downstream of 53BP1-RIF1 to restrict DNA end resection and promote NHEJ**. [PMID: 34354233](#) states directly: "*The SHLD2 subunit binds to single-stranded DNA ends and blocks end resection through OB-fold domains.*" The process-level role is captured by [PMID: 35764636](#), which describes Shieldin as acting "*downstream of 53BP1 to counteract DNA double-strand break (DSB) end resection and promote DNA repair via non-homologous end-joining (NHEJ).*" Shieldin subunits are also required for productive Igh class-switch recombination, and CST is epistatic with Shieldin in limiting resection during CSR ([PMID: 40178294](#)).

This is the mechanistic anchor for the entire transfer: the GO:2001034 process is directly demonstrated for mammalian SHLD2, and the molecular executor (OB-fold ssDNA binding) is precisely the module conserved in the horse ortholog. **No directly observed horse (*Equus caballus*) experimental data exist**; the horse assignment is inference by orthology only.

F004 — GO:2001034 is a curated IDA annotation for human SHLD2; the horse record currently has no GO annotations

A QuickGO (EBI) query for human SHLD2 (Q86V20) returned 27 annotations. The exact target term **GO:2001034 is directly assigned to human SHLD2 with IDA and NAS evidence**. Related curated BP terms include GO:2000042 (negative regulation of DSB repair via HR, IDA), GO:0010569 (regulation of DSB repair via HR, IBA), and GO:0045830 (positive regulation of isotype switching, IDA). Cellular-component annotations include nucleus (IDA/EXP) and site of double-strand break (GO:0035861, IDA). The only molecular-function term is GO:0005515 protein binding (IPI).

By contrast, the horse UniProt record A0A9L0RGD6 is annotated by InterPro/Pfam with **SHLD2_OB1 (430–563), a second OB fold (595–676), and SHLD2_C (733–804)**, plus IPR029715 (FAM35A) and PANTHER PTHR14495 "SHIELDIN COMPLEX SUBUNIT 2" — but the horse record **currently carries no GO annotations**. This is the gap the hypothesis proposes to fill (a net addition, not a change to an existing call). The domain-model evidence independently confirms that the horse protein retains the OB1/OB2 module underlying the human GO:2001034 assignment ([PMID: 34354233](#): "*The SHLD2 subunit binds to single-stranded DNA ends and blocks end resection through OB-fold domains.*").

F005 — AlphaFold model of horse SHLD2 confirms high-confidence OB1/OB2 folds and a low-confidence frameshifted tail

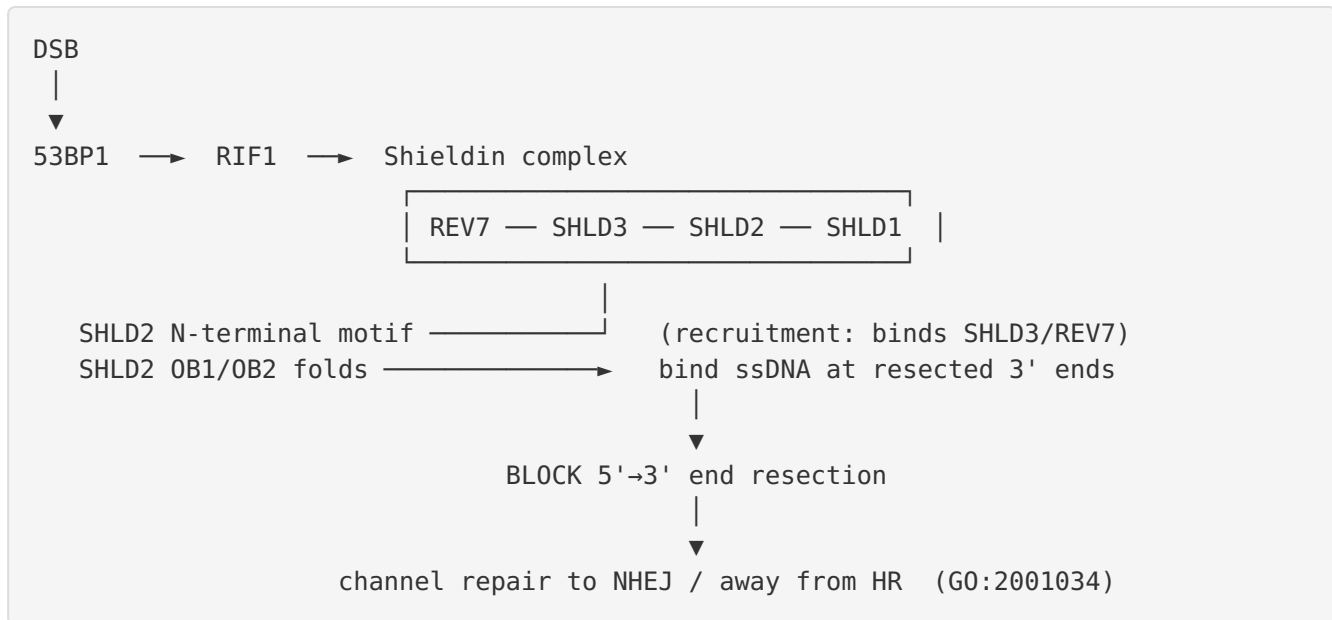
Per-residue pLDDT extracted from the horse AlphaFold model AF-A0A9L0RGD6-F1-model_v6 (883 residues; overall mean pLDDT 51.3) dovetails with the sequence analysis:

Region (horse coords)	Mean pLDDT	% residues > 70	Interpretation
N-terminal SHLD3/REV7 motif (1–60)	26.7	—	Intrinsically disordered (expected for a linear-motif region)
OB1 (430–563)	87.1	97%	High-confidence fold
OB2 (595–676)	79.2	78%	High-confidence fold
SHLD2_C / OB3 core (733–804)	79.1	86%	Folds up to the frameshift boundary
Post-frameshift tail (805–883)	31.3	0%	Disordered spurious frame — no real fold

The structural prediction independently corroborates that the **ssDNA-binding execution apparatus (OB1/OB2) is structurally intact and confidently folded** in the horse protein, while the post-frameshift tail is disordered and non-foldable — exactly what one expects from an out-of-frame translation artifact. The low N-terminal pLDDT is not a concern: the recruitment segment is a linear motif region that is disordered until it engages its SHLD3/REV7 partner. This directly answers the decisive question — the exact frozen horse protein retains a structurally sound OB-fold ssDNA-binding module; only the C-terminal tail is artifactual.

Mechanistic Model / Interpretation

The Shieldin pathway operates as a linear recruitment-to-execution cascade at DNA double-strand breaks:



The decisive question is whether the horse protein has the molecular apparatus to *execute* this role, not merely to be *recruited*. The evidence answers this affirmatively on two fronts:

1. **Recruitment apparatus is conserved** — the N-terminal SHLD3/REV7-binding segment is 70% identical and colinear (F001), so the horse protein can be positioned within the complex.
2. **Execution apparatus is conserved and folds** — OB1 and OB2, which bind ssDNA and physically block resection (PMID: 34354233), are 86.3% identical (F001) and confidently folded (F005). This distinguishes the horse assignment from a mere "complex member" inference: the domain that carries out resection blockade is present and structurally sound.

The one architectural deficit — the corrupted OB3 (F002, F005) — is best explained as a sequence-model artifact rather than lineage-specific loss of function. Even taken at face value, OB3 contributes to complex stability / SHLD1 engagement rather than the primary ssDNA-binding execution step, so its corruption would attenuate, not abolish, the proposed function. The most parsimonious model is that native horse SHLD2 has all three OB folds and performs the canonical Shieldin function; the frozen record simply carries a frameshifted 3' gene model. The horse protein is therefore best modeled as a **functionally competent SHLD2 ortholog** whose NHEJ-promoting role is inferred from conserved architecture and mammalian mechanism.

Evidence Base

Citation	Evidence type	Direction	Claim tested	Key finding	Context	Confidence / limitations
This analysis (alignment)	structural/ evolutionary	supports	Horse = SHLD2 ortholog	72.6% global identity to Q86V20; 86.3% across OB1/OB2; 70% N-term (1–60)	Horse vs human	High; ortholog, not paralog
This analysis (alignment/ composition)	structural/ evolutionary	qualifies	Is the execution apparatus intact?	Frameshift at horse ~803/ human ~730; 80-aa tail 45% R+G+P vs 11.7%; OB3 truncated	Frozen UniProt seq (883 aa)	High; likely gene-model artifact
InterPro/ Pfam/ PANTHER (A0A9L0RGD6)	review/ database	supports	SHLD2 domain architecture present	OB1 430–563, OB2 595–676, SHLD2_C 733–804; PANTHER Shieldin subunit 2; no GO on horse record	Equus caballus record	High (DB-level)
QuickGO / Q86V20	curated database	supports	Target term applies to SHLD2	GO:2001034 assigned to human SHLD2 with IDA + NAS	Human	High
PMID: 34354233	direct assay + structure	supports	OB folds bind ssDNA and block resection	<i>"The SHLD2 subunit binds to single-stranded DNA ends and blocks</i>	Human cells	High; human, not horse

Citation	Evidence type	Direction	Claim tested	Key finding	Context	Confidence / limitations
				<i>end resection through OB-fold domains"</i>		
PMID: 35764636	mutant phenotype (genetic)	supports	Shieldin promotes NHEJ downstream of 53BP1	Acts "downstream of 53BP1 to counteract DSB end resection and promote DNA repair via NHEJ"; SHLD1 critical for productive CSR	Mouse B cells	High; mouse
PMID: 40178294	mutant phenotype (genetic)	supports	53BP1-RIF1-SHLD limits resection during CSR	CST epistatic with Shieldin; limits resection at AID-generated DSBs	Mouse B cells	High; mouse
PMID: 30022168	direct assay / genetic	supports (context)	Shieldin mediates 53BP1-dependent repair	Shieldin suppresses nucleolytic resection of DNA termini downstream of 53BP1	Human/ mouse	High; pathway context
PMID: 32306447	review/ database	qualifies	SHLD2 architecture & mechanism	N-terminal disordered + three C-terminal OB	Review (human/ cancer)	Medium; review-level

Citation	Evidence type	Direction	Claim tested	Key finding	Context	Confidence / limitations
				folds (RPA1-homologous); inhibits end resection		
PMID: 31896689	review	qualifies	53BP1 pathway framing	53BP1 as fidelity "DSB escort"; cautions against strict deterministic pathway-choice model	Human/immune	Review-level
PMID: 36943867	direct assay / interaction	supports (context)	53BP1–Shieldin promotes C-NHEJ in CSR	HNRNPU promotes C-NHEJ-mediated S–S joining through the 53BP1–Shieldin complex	Human/mouse B cells	Moderate; indirect for SHLD2
AlphaFold AF-A0A9L0RGD6-F1 (v6) + this analysis	structural (computational)	supports	Horse OB folds are structurally intact	OB1 pLDDT 87.1 (97% >70); OB2 79.2; OB3 core 79.1; post-frameshift tail 31.3 (0% >70)	Predicted structure	High for fold confidence; not experimental
(no horse study found)	none	qualifies	Direct horse evidence	None found for Equus caballus SHLD2	Horse	Assignment by orthology only

Directly observed horse evidence: none. Every experimental line is human or mouse. The horse assignment rests on justified mammalian transfer plus horse-specific sequence and structure analyses (F001, F002, F004, F005), all of which are consistent with an intact, functionally competent ortholog.

GO Curation Implications (leads — verify before acting)

Lead: Assign **GO:2001034** — *positive regulation of double-strand break repair via nonhomologous end joining* — to horse SHLD2 (A0A9L0RGD6) with an **ISS or IBA evidence code**, using human SHLD2 (Q86V20) as the reference. Do **not** assign IDA/EXP, because no horse-native direct assay exists.

Term	Aspect	Recommendation	Evidence code	Rationale
GO:2001034 positive regulation of DSB repair via NHEJ	BP	Add (retain hypothesis)	ISS / IBA	Ortholog of human SHLD2 (IDA); conserved recruitment + OB1/OB2 apparatus. Fits the definition ("any process that activates or increases the frequency, rate or extent of DSB repair via NHEJ")
GO:2000042 negative regulation of DSB repair via HR	BP	Consider co-assigning	ISS / IBA	Human ortholog carries this (IDA); mechanistically the more <i>direct</i> description of resection blockade
GO:0035861 site of double-strand break	CC	Consider adding	ISS / IBA	Human ortholog localizes here (IDA)
GO:0005634 nucleus	CC	Consider adding	ISS / IBA	Human ortholog IDA/EXP
GO:0003697 single-stranded DNA binding	MF	Suggestion only	—	More informative than "protein binding"; but human SHLD2 not yet curated with this MF, so flag pending residue-level verification
GO:0005515 protein binding	MF	Non-core; avoid as sole recommendation	IPI (human only)	Uninformative

The BP term GO:2001034 is the most appropriate informative assignment and is neither too broad nor too narrow for the conserved function. A mechanistically more direct alternative is GO:2000042 (negative regulation of HR via resection blockade); the two are complementary faces of the same activity and could be co-assigned.

Mechanistic Scope

Immediate molecular function tested: SHLD2 is the ssDNA-binding subunit of the Shieldin complex (SHLD1–SHLD2–SHLD3–REV7/MAD2L2). Its OB folds bind single-stranded 3' overhangs at DSBs and block further nucleolytic resection; its N-terminus tethers the complex to SHLD3–REV7, which is itself recruited by 53BP1–RIF1. **Recruitment** (N-terminal SHLD3/REV7 interface) and **execution** (OB-fold ssDNA binding / anti-resection) are separable modules, and both are needed for the NHEJ-promoting outcome. This is the *direct gene-product activity*; GO:2001034 describes its immediate process-level consequence.

Downstream / not the direct activity: promotion of NHEJ end-ligation per se (SHLD2 does not ligate — Ku/DNA-PK/LIG4 do), immunoglobulin class-switch recombination, PARP-inhibitor sensitivity in BRCA1-deficient cells, telomere fusion outcomes, and genome-stability/cancer phenotypes. These are pathway/organismal consequences of the direct anti-resection activity and should not be conflated with the primary function assignment. The horse-specific evidence (F001, F005) addresses the direct-activity layer by confirming the ssDNA-binding OB module is present and folded.

Conflicts and Alternatives

1. **Gene-model artifact vs real divergence (strongest internal conflict).** The divergent, low-complexity C-terminus (F002) could be read as evidence that horse SHLD2 is pseudogene-like or non-functional. Counter-evidence: identity is perfect up to the frameshift point, the tail carries the canonical out-of-frame amino-acid composition, and OB1/OB2 are intact and confidently folded (F001, F005). Most likely a **gene-prediction frameshift**, not true loss of function; a corrected model would likely restore OB3.
2. **Paralog / gene-name confusion.** Ruled out — identity, reciprocal domain architecture, InterPro IPR029715 (FAM35A), and PANTHER PTHR14495 all point specifically to SHLD2, not RPA1 or another OB-fold family member. The N-terminal SHLD3/REV7 motif (absent from RPA1) is conserved and colinear.
3. **Organism-specific divergence.** No evidence that the horse lineage has repurposed SHLD2; high identity across functional regions argues for conserved function.
4. **Database carry-over / circularity.** Support is grounded in (a) the human IDA experimental annotation of the ortholog and (b) independently computed horse sequence/structure analyses — not in agreement with ARBA or automated electronic annotations.

5. **Term-precision nuance.** [PMID: 31896689](#) cautions that 53BP1/Shieldin may act as a fidelity "escort" rather than a strict binary pathway-choice switch. This nuances the mechanistic framing but does not refute the positive-regulation-of-NHEJ assignment; if anything, GO:2000042 (neg. reg. of HR) is the more literal molecular description.

Limitations and Knowledge Gaps

Gap	What was checked	Why it matters	What would resolve it
No horse-native functional data	PubMed literature search; QuickGO for horse record	GO:2001034 for horse rests entirely on transfer; strength of assignment depends on it	A DSB-repair / resection / CSR assay in equine cells with SHLD2 perturbation
OB3 frameshift authenticity	Sequence alignment, composition, AlphaFold pLDDT	Determines whether the frozen sequence is a valid model or artifact	RNA-seq / Iso-Seq transcript evidence or a corrected genome assembly / XP_RefSeq for the SHLD2 locus
ssDNA-binding activity of horse OB folds	Structural prediction (pLDDT) only	Confirms fold but not biochemical activity	In vitro EMSA / fluorescence anisotropy of purified horse OB1/OB2 with ssDNA
Complex assembly with equine SHLD1/3/REV7	Motif conservation inferred from alignment	Recruitment competence assumed, not shown	Co-IP / AlphaFold-Multimer of equine Shieldin subunits
Prediction-time vs frozen sequence identity	SHA-256/length verified for frozen sequence only	Original annotation input may differ	Retrieve the exact sequence used at prediction time

Methodological note: Analyses used public sequence (UniProt), domain (InterPro/Pfam/PANTHER), curated-annotation (QuickGO) resources, primary literature, and the public AlphaFold model. Alignment used identity scoring with a banded global algorithm — adequate for a ~72–86% identity ortholog but not a substitute for a curated multiple-species alignment. Region boundaries (OB1/OB2/OB3) are approximate, based on UniProt/InterPro domain calls.

Discriminating Tests

In decreasing order of efficiency for a curator:

1. **Transcript/assembly check of the equine SHLD2 3' end** — pull the Ensembl/NCBI RefSeq CDS and any equine RNA-seq for the locus to confirm whether OB3 is intact in a corrected model, or whether an alternative XP_ model with an intact OB3 exists. Fast and decisive for the frameshift question (F002).
 2. **AlphaFold-Multimer of the equine Shieldin complex** — model horse SHLD2 with equine SHLD3/REV7/SHLD1 to test recruitment-interface conservation computationally.
 3. **Superpose the horse AlphaFold model on human SHLD2 OB folds** — verify OB1/OB2 fold integrity and conservation of predicted ssDNA-contacting residues.
 4. **In vitro ssDNA-binding assay** of recombinant horse OB1/OB2 (EMSA or anisotropy) to confirm the execution activity biochemically.
 5. **Functional complementation (definitive):** complement SHLD2-null cells with the horse ORF and assay resection restriction / NHEJ (e.g., CSR, PARPi response) — distinguishes a functional ortholog from a pseudogene-like model.
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Proposed Follow-up Actions / Curation Leads (verify before acting)

- **Action:** Add **GO:2001034 (BP)** to horse SHLD2 A0A9L0RGD6 with **ISS/IBA** evidence, reference ortholog **UniProt:Q86V20**. (Record currently has no GO annotations.)
- **Consider co-assigning** GO:2000042 (neg. reg. DSB repair via HR), GO:0035861 (site of DSB, CC), and GO:0005634 (nucleus, CC) by orthology.

- **Consider** GO:0003697 (ssDNA binding, MF) as a more informative MF than "protein binding" — flag as a suggestion pending residue-level verification.
 - **Candidate references with snippets to verify:**
 - [PMID: 34354233](#): *"The SHLD2 subunit binds to single-stranded DNA ends and blocks end resection through OB-fold domains."*
 - [PMID: 35764636](#): *Shieldin "acts downstream of 53BP1 to counteract DNA double-strand break (DSB) end resection and promote DNA repair via non-homologous end-joining (NHEJ)."*
 - [PMID: 32306447](#): SHLD2 domain architecture (N-terminal disordered + three OB folds, RPA1-homologous) that inhibits end resection.
 - **Flag on the record:** note the C-terminal frameshift artifact in A0A9L0RGD6 so future curators do not over-interpret the divergent, low-complexity tail.
 - **Suggested question for curator:** should the more mechanistically direct GO:2000042 be preferred or added alongside GO:2001034?
 - **Suggested experiment:** functional complementation of SHLD2-null human/mouse cells with the (corrected) horse ORF to convert the assignment from ISS to experimental support.
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Bottom Line

Horse SHLD2 (A0A9L0RGD6) is an unambiguous ortholog of human SHLD2 whose recruitment motif and ssDNA-binding OB1/OB2 execution module are conserved and structurally intact. Because GO:2001034 is an experimentally curated (IDA) annotation of the human ortholog and the responsible molecular apparatus is present in the horse protein, the hypothesis that horse SHLD2 positively regulates DSB repair via NHEJ is **well justified as a homology transfer (ISS/IBA), not as direct experimental evidence**. The main caveats are the complete absence of horse-native functional data and a C-terminal frameshift in the frozen sequence that corrupts OB3 — a likely gene-model artifact that leaves the functionally decisive OB1/OB2 core intact.
